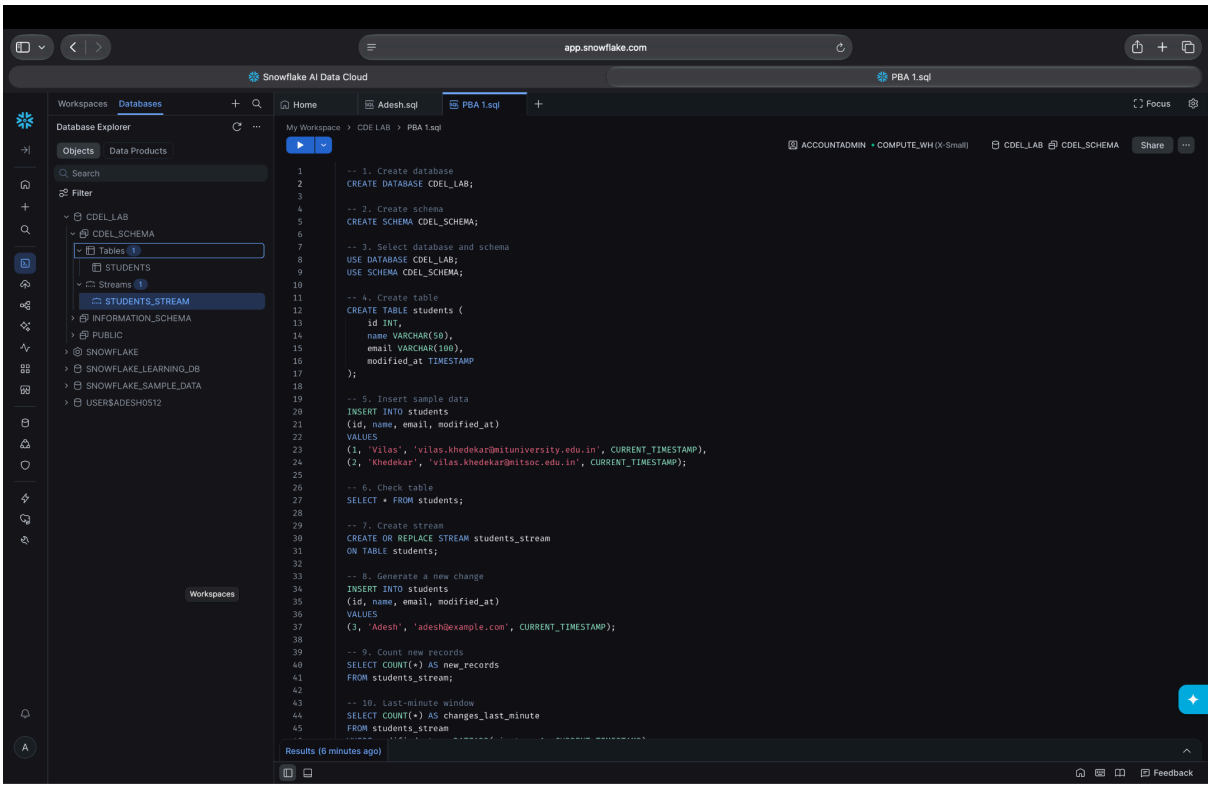
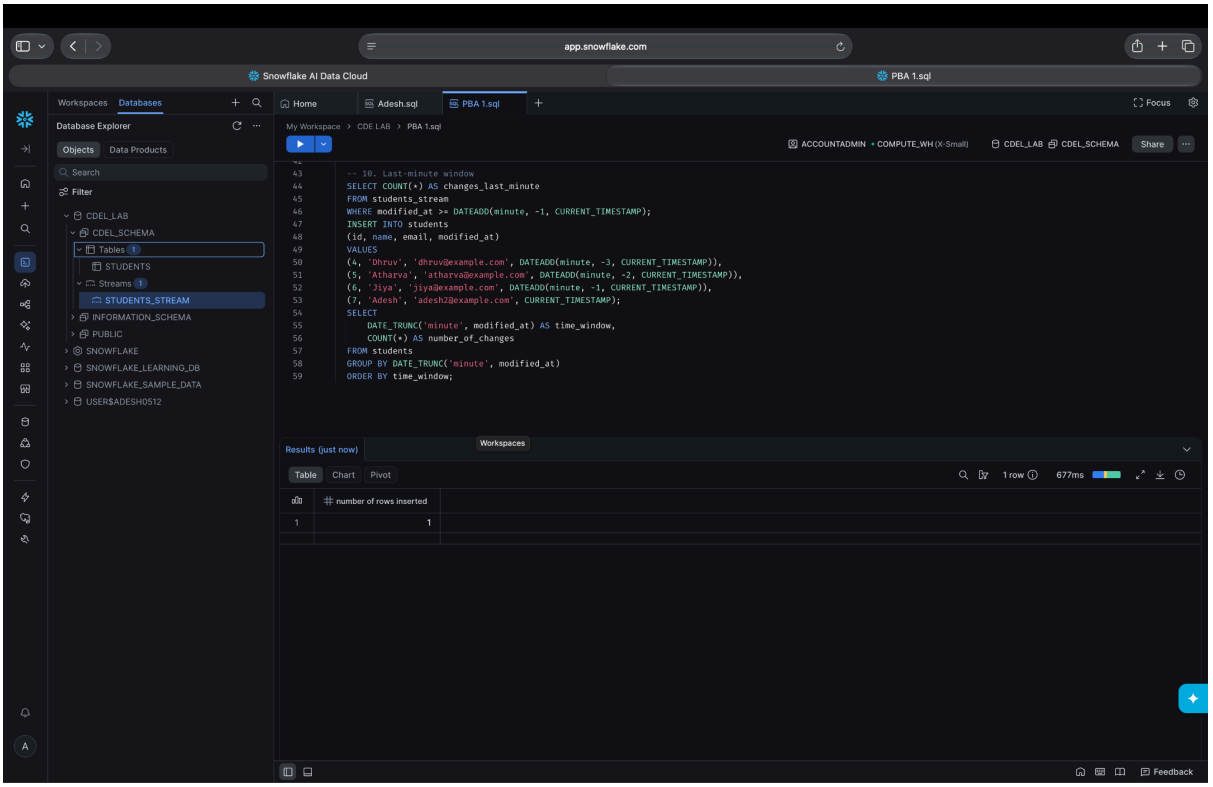
NAME : Adesh Deepak Patankar
CLASS : TY-BDCE

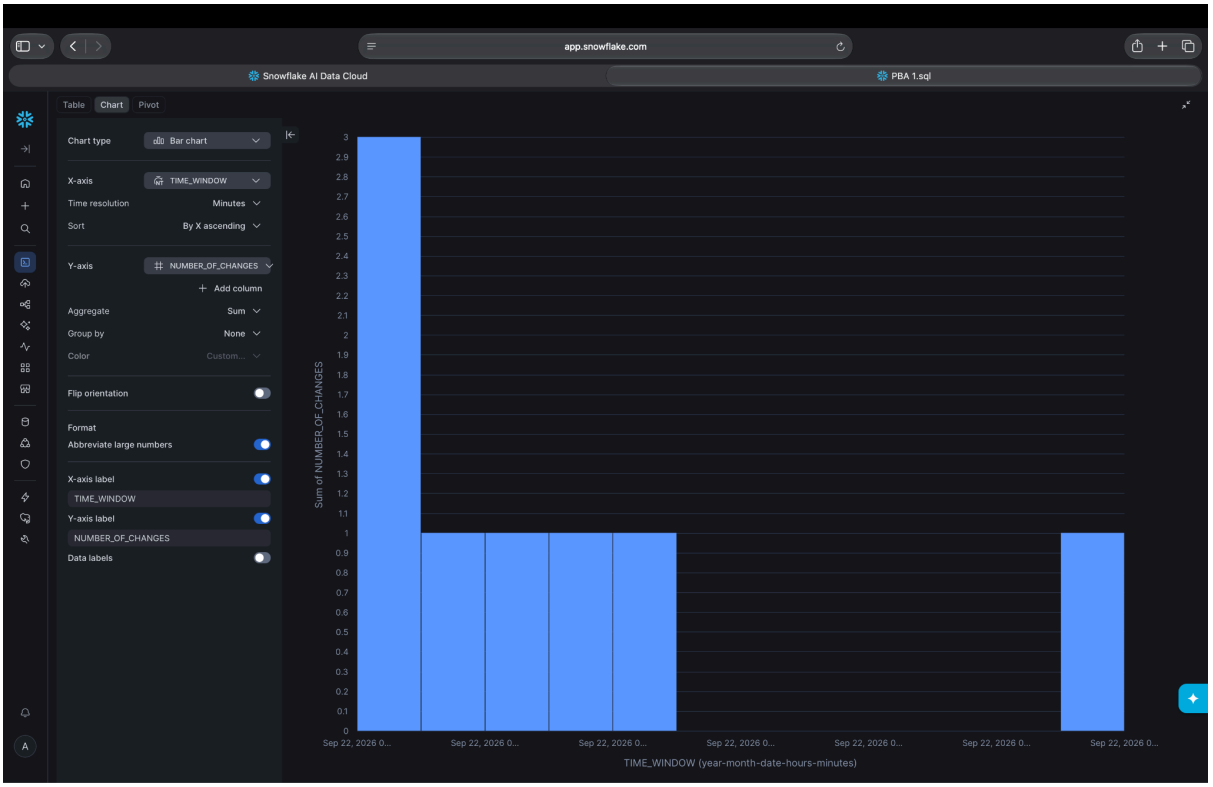
SUBJECT : CDE - LAB
PBA - 1

Problem Statement :- Implement Real-time analytics (ex : counting, windowing) on the streaming data and visualize the results.

SCREENSHOTS OF THE PERFORMED EXPERIMENT




app.snowflake.com
Snowflake AI Data Cloud
PBA 1.sql
Workspaces
Databases
Database Explorer
Objects
Data Products
Search
Filter
CDEL_LAB
CDEL_SCHEMA
Tables 1
STUDENTS
Streams 1
STUDENTS_STREAM
INFORMATION_SCHEMA
PUBLIC
SNOWFLAKE
SNOWFLAKE_LEARNING_DB
SNOWFLAKE_SAMPLE_DATA
USER$ADESH0512
Home
Adesh.sql
PBA 1.sql
Focus
My Workspace › CDE LAB › PBA 1.sql
ACCOUNTADMIN • COMPUTE_WH (X-Small)
CDEL_LAB CDEL_SCHEMA
Share
-- 10. Last-minute window
SELECT COUNT(*) AS changes_last_minute
FROM students_stream
WHERE modified_at >= DATEADD(minute, -1, CURRENT_TIMESTAMP);
INSERT INTO students
(id, name, email, modified_at)
VALUES
(4, 'Dhruv', 'dhruv@example.com', DATEADD(minute, -3, CURRENT_TIMESTAMP)),
(5, 'Atharva', 'atharva@example.com', DATEADD(minute, -2, CURRENT_TIMESTAMP)),
(6, 'Jiya', 'jiya@example.com', DATEADD(minute, -1, CURRENT_TIMESTAMP)),
(7, 'Adesh', 'adesh2@example.com', CURRENT_TIMESTAMP);
SELECT
DATE_TRUNC('minute', modified_at) AS time_window,
COUNT(*) AS number_of_changes
FROM students
GROUP BY DATE_TRUNC('minute', modified_at)
ORDER BY time_window;
Results (just now)
Workspaces
Table
Chart
Pivot
1 row
677ms
number of rows inserted
1
1
Feedback


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Snowflake AI Data Cloud
PBA 1.sql
Table
Chart
Pivot
Chart type
Bar chart
X-axis
TIME_WINDOW
Time resolution
Minutes
Sort
By X ascending
Y-axis
NUMBER_OF_CHANGES
Add column
Aggregate
Sum
Group by
None
Color
Custom...
Flip orientation
Format
Abbreviate large numbers
X-axis label
TIME_WINDOW
Y-axis label
NUMBER_OF_CHANGES
Data labels
Sum of NUMBER_OF_CHANGES
TIME_WINDOW (year-month-date-hours-minutes)
Sep 22, 2026 0...



CONCLUSION : In this experiment, a Snowflake Stream was created to track changes in the students table. Counting and time-based windowing were performed on the streaming data, and the results were visualized.